MILITARY SPECIFICATION
BAND, OBTURATING, FOR
175MM, PROJECTILE

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. Scope

1.1 This specification covers one type of obturating band for use on the 175MM artillery projectile.

2. Applicable documents

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

9211857	-	Band, Obturating

FSC: 1320

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(Copies of specifications, standards, drawings and publication required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The material shall be in accordance with applicable drawings and specifications.

3.2 Obturating bands. The obturating bands shall comply with all requirements specified on Drawing (dwg.) 9211857. The demensions of the finished bands shall apply after moisture conditioning, and after the bands have cooled to room temperature.

3.2.1 Moisture content. The obturating bands to be subjected to examination test shall be moisture conditioned until 2.0 to 3.0 percent, by weight, moisture content is attained (see 6.4). Testing shall be as specified in 4.4.1.

3.3 Tensile strength. The obturating bands shall have a tensile strength of 6,000 pounds per square inch (psi), minimum (min.) when tested as specified in 4.4.2. The tensile strength shall apply to bands which have been moisture conditioned.

3.4 Workmanship. All obturating bands shall be fabricated and finished in a thorough, workmanlike manner and all manufacturing and processing operations shall be correctly performed. The bands shall be clean and free of flash, air bubbles, holes, blisters, cracks, dirt, grease and other foreign matter.

3.5 First Article Inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Govern-

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ment. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five bands which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to

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the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspection to be performed. Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. If any assembly or components fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly to component to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawings, same drawing revisions same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Plastic molding material from one manufacturer.

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4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Band, obturating (see dwg. 9211857).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Inside diameter, maximum	Gage
Minor:	AQL 4.0 percent	
201.	Inside diameter, min	Gage
202.	Thickness	Gage
203.	Width	Gage
204.	Diameter of holes	Gage
205.	Location of holes	Gage
206.	Number of holes incorrect	Visual
207.	Radius or chamfer missing or incorrect	Visual
208.	Evidence of poor workmanship (see 3.4)	Visual

4.3.3 Testing

4.3.3.1 Tensile strength (Major defect) (see 3.3). Ten obturating bands shall be moisture conditioned and tested from each lot containing less than 5,000 bands. For lots containing 5,000 or more bands, 10 bands shall be moisture conditioned and tested for each 5,000 bands or portion thereof produced. Failure

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of 2 or more bands to comply with the requirement shall be cause for rejection of the bands represented by the sample. If one band fails, an additional 10 bands shall be conditioned and tested. Failure of any of these additional bands shall be cause for rejection of the bands represented by the sample. Testing shall be performed as specified in 4.4.2.

4.3.4 Inspection equipment. Inspection equipment required to perform the inspection prescribed in this specification is identified in the Examination 4.3.2 and Test Method 4.4, Paragraphs herein. See 6.2 for details concerning responsibilities for inspection equipment design approval.

4.4 Test Methods and Procedures

4.4.1 Moisture content. Those obturating bands which are to be moisture conditioned (see 6.4) shall be weighed on an analytic balance and then heated in an air-circulating oven for 24 hours at a temperature of 105 degrees centigrade (°C). The bands shall then be cooled to room temperature in a desicator and reweighed. The bands shall then be heated for one hour at 105°C, cooled, and reweighed. The procedure of heating for one hour, cooling, and reweighing shall be continued until the bands reach a constant weight. The difference between the initial weight and the final constant weight shall be taken as the amount of moisture originally present in the band. The moisture content (percent) shall be calculated as follows:

$$MC = \frac{W_o - W_2}{W_o} (100)$$

Where:

MC = Moisture content (percent)
W_o = Initial weight
W_2 = Final constant weight

4.4.2 Tensile strength. A fixture shall be employed which consists of two semicircular steel discs with a diameter slightly less than the inside diameter of the obturating band to be tested and a thickness at least that of the band. The fixture shall be placed together to form a round disc and the band fitted around the fixture. The discs shall then be attached to the self-aligning grips of a tensile testing machine by means of two suitable

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harnesses. Bands tested shall have the holes positioned as closely as practicable at right angles to the direction of pull. With the band and fixture in place, the tensile testing machine shall be operated at a constant rate of 2.0 inches per minute until the band breaks. The tensile strength shall be calculated as follows:

$$S_t = \frac{P}{2\ (A_b - 2A_h)}$$

Where:

S_t = Tensile strength
P^t = Load at failure of the band
A_b = Cross-sectional area of the band
A_h = Cross-sectional area of one hole (hole diameter times thickness of band)

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level C. The obturating bands shall be packed in accordance with MIL-STD-1169.

5.2 Marking. The marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167.
c. First Article Inspection.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the con-

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tractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any specific fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification, paragraph, and defect numbers from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted

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for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-A, Dover, N.J. in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 It is noted that the size of inspection lots of components assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.4 Obturating bands which are to be examined (see 4.3.2) or tested (see 4.3.3) shall be moisture conditioned by immersion in boiling water until the required moisture content is attained (see 3.2.1). In addition, two extra bands per conditioning batch shall be moisture conditioned. These additional two bands shall be tested in accordance with 4.4.1 as a check on the moisture content of the batch from which they were taken. Moisture conditioning batches which fail to meet the moisture content requirement shall be reconditioned until the requirement is met.

Custodian:
ARMY-PA

Preparing Activity
ARMY-PA

Project Number: 1320-A222

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OMB Approval
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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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☐ YES ☐ NO *(If "Yes", in what way?)*

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